M.Sc. Thesis Overview

Simulation and Optical Design of Multilayer Transparent Conducting Oxides

SIMULATION AND OPTICAL DESIGN OF MULTILAYER TRANSPARENT CONDUCTING OXIDES

M.Sc. Physics Thesis Overview
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PURPOSE OF THIS WEB COPY

This text-only PDF is a concise overview prepared for the online research portfolio. It presents the central motivation, modelling approach, results, limitations and future work of the full M.Sc. thesis. The original submitted thesis contains the complete figures, tables, equations and bibliography.

ABSTRACT

The thesis investigates the optical design of an indium zinc oxide / silver / indium zinc oxide (IZO/Ag/IZO) multilayer as a transparent conducting electrode for photovoltaic devices. Conventional transparent conducting oxides face a trade-off between optical transmission and electrical conductivity. An oxide-metal-oxide structure can use an ultrathin metal layer for lateral conduction while the surrounding dielectric layers are designed to improve optical transmission through interference and admittance matching.

SETFOS simulations were used to study transmission, reflection and absorption while systematically varying the thickness of the IZO and silver layers. Python-based analysis was used to calculate wavelength-averaged transmission, transmitted photon flux and theoretical short-circuit-current limits under the AM1.5G spectrum. The work also examined sheet resistance and Haacke’s figure of merit to avoid treating optical transmission as the only design criterion.

A central part of the thesis was the treatment of ultrathin silver. Bulk optical constants are not automatically valid for sub-10 nm films because the metal may grow as islands before forming a continuous layer. Thickness-dependent behaviour was interpreted using a Drude-based correction together with literature on Volmer-Weber growth, localized surface plasmon resonance and wetting-layer-assisted continuity.

The idealised simulations identified IZO(45 nm)/Ag(8 nm)/IZO(60 nm) for the 300-800 nm range and IZO(45 nm)/Ag(8 nm)/IZO(40 nm) for the 300-1300 nm range as promising optical designs. These simulated improvements are theoretical design limits, not direct predictions of fabricated-device performance. Real structures introduce surface roughness, thickness non-uniformity, scattering, grain boundaries, interface diffusion, imperfect film continuity and differences between assumed and measured optical constants.

1. RESEARCH MOTIVATION

Transparent conducting electrodes must transmit light while carrying current. Increasing oxide thickness may reduce sheet resistance but can also increase optical loss. Reducing thickness can improve some optical properties while weakening lateral conductivity. An oxide-metal-oxide stack offers another design space: a thin metal layer supports conduction, and the upper and lower oxide layers modify the optical field.

The project initially began as a transparent-electrode study for tandem solar cells. During the work, the emphasis shifted increasingly toward photonics and light-matter interaction. The questions that became most engaging were how nanoscale thickness changes reshape the spectrum, how the optical constants of ultrathin materials differ from bulk values, how fields are distributed inside a multilayer, and how an ideal optical model should be connected to fabrication constraints.

2. OPTICAL DESIGN APPROACH

The optical response of the multilayer was modelled using SETFOS. Thickness sweeps were carried out for the top IZO, intermediate Ag and bottom IZO layers. The following quantities were considered:

- Spectral transmission, reflection and absorption.
- Average transmission in wavelength ranges relevant to single-junction and tandem devices.
- AM1.5G-weighted transmitted photon flux.
- Theoretical short-circuit-current limit, used as an optical comparison metric.
- Approximate sheet resistance of the multilayer.
- Haacke’s figure of merit, combining transparency and conductivity.

Python was used for data processing and comparison between candidate structures. The workflow was designed to identify optical mechanisms and promising thickness combinations rather than to claim that ideal simulated gains would be reproduced unchanged after fabrication.

3. ULTRATHIN SILVER AND NON-BULK OPTICAL RESPONSE

Silver films below approximately 10-15 nm can be discontinuous. During early growth, deposited atoms may form separated islands. As thickness increases, the islands connect and eventually form a continuous conducting film. This transition affects both optical and electrical behaviour.

The thesis therefore did not treat bulk silver constants as universally valid. A Drude-based approach was used to explore thickness-dependent damping and optical response, supported by literature on island growth, percolation, localized plasmon resonances and wetting layers. A thin germanium wetting layer was considered in relation to improved continuity of ultrathin silver.

This analysis also highlighted an important experimental point: a nominal thickness does not uniquely determine performance. Deposition rate, substrate condition, vacuum quality, surface energy, wetting layer, roughness and post-deposition history can all alter the morphology and therefore the measured optical constants.

4. SIMULATED DESIGNS AND INTERPRETATION

For the 300-800 nm wavelength range, the simulation identified an IZO(45 nm)/Ag(8 nm)/IZO(60 nm) structure as a strong candidate. Relative to the simulated 140 nm IZO reference, the model indicated an 8.55 percent increase in the theoretical short-circuit-current cap.

For the broader 300-1300 nm range, an IZO(45 nm)/Ag(8 nm)/IZO(40 nm) design gave a simulated 13.06 percent improvement in wavelength-averaged transmission and a 6.17 percent improvement in the theoretical current cap relative to the chosen reference.

These percentages describe the output of an idealised optical comparison. They should not be interpreted as guaranteed gains after deposition or as direct efficiency improvements in a complete solar cell. Fabricated films typically show smaller gains because the real stack differs from the ideal model. Important causes include:

- Incomplete or non-uniform silver coverage.
- Surface and interface roughness.
- Grain-boundary and defect-related electrical losses.
- Thickness variation across the substrate.
- Scattering and absorption not fully represented by the optical data.
- Interdiffusion and chemical changes at interfaces.
- Differences between literature optical constants and the deposited films.
- Additional parasitic absorption in a complete device stack.

The value of the simulation is therefore to reveal optical headroom, identify sensitive parameters and reduce the experimental search space. Measured optical constants, morphology and device data should be returned to the model in later design iterations.

5. ELECTRICAL AND FIGURE-OF-MERIT ANALYSIS

Optical transmission alone is insufficient for evaluating a transparent electrode. The silver layer can strongly reduce lateral resistance, but only if the film is sufficiently continuous. The thesis compared approximate sheet resistance with optical performance and evaluated Haacke’s figure of merit.

The simulated optimized stacks showed a two-to-fourfold enhancement in the figure of merit compared with the selected single-layer reference under the assumptions used. As with the optical results, this represents model-based potential. Experimental sheet resistance depends strongly on morphology, contact quality and the electrical behaviour of the interfaces.

6. DEPOSITION-METHOD ANALYSIS

The thesis compared thermal evaporation and magnetron sputtering for ultrathin silver deposition. A geometric material-utilisation estimate suggested that sputtering could require approximately 24 times less target mass per deposition than the selected evaporation configuration because of reduced angular loss and improved confinement of the deposited flux.

This calculation is a tool-specific approximation, not a universal ratio for all evaporation and sputtering systems. Actual utilisation depends on source geometry, target size, throw distance, shielding, deposition area and operating conditions. The comparison nevertheless showed why process efficiency and scalability should be considered alongside optical performance.

7. EXPERIMENTAL FOUNDATION

The research was supported by hands-on experience with RF magnetron sputtering, metal evaporation, cleanroom processing, vacuum handling, UV-Vis-NIR spectroscopy, four-point-probe measurements and profilometry. This experimental exposure informed the interpretation of the simulations and made clear why ideal layer thicknesses must be tested against process capability and film morphology.

8. MAIN LESSONS

- Light management in a transparent electrode is a coupled optical, electrical and materials

problem.
- Ultrathin-metal optical constants must be treated carefully; bulk data can be misleading.
- A large simulated improvement should be presented as theoretical optical potential, not promised fabricated-device performance.
- The best workflow is iterative: design, fabricate, measure, update the model and redesign.
- Interface quality and morphology are part of the optical problem, not secondary fabrication details.
- A useful design must be evaluated for manufacturability, material efficiency and device compatibility.

9. FUTURE WORK

The next step is physical fabrication and systematic characterization of the optimized stacks. Future work should include:

- Deposition of controlled IZO/Ag/IZO thickness series.
- Ellipsometric extraction of optical constants from the actual deposited layers.
- UV-Vis-NIR transmission and reflection measurements.
- Four-point-probe sheet-resistance measurements.
- AFM or electron-microscopy investigation of silver continuity and roughness.
- Comparison of wetting layers and deposition conditions.
- Integration into complete photovoltaic structures.
- Recalibration of the optical model using measured morphology and optical constants.

10. CONCLUSION

The thesis established a fabrication-aware optical-design workflow for IZO/Ag/IZO transparent conducting multilayers. It identified promising thickness combinations and clarified the physical assumptions behind their simulated performance. More importantly, the work developed a research perspective in which modelling and fabrication are not separate stages. Simulations reveal mechanisms and guide experiments; experiments determine which assumptions are realistic and provide the data needed to improve the next model.

The project also marked a shift in research interest from photovoltaic electrode optimisation toward the broader physics of light-matter interaction, multilayer photonics and nanostructured optical materials.